Theoretical Framework: Unified Frequency Aggregation System

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1 Introduction

This document presents a conceptual model for a high-efficiency energy extraction system. The architecture is based on the premise that an array of distributed receivers, all synchronized to a common frequency, can collectively aggregate power to achieve a high-density output.

2 Synchronized Resonant Architecture

The system operates on the principle of distributed resonance. Each receiver unit (i) is precisely tuned to the transmission frequency (f_r), ensuring maximum capture capacity:

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

Where L and C are the inductance and capacitance components of the receiver circuit, optimized for perfect harmonic alignment.

3 Cumulative Power Aggregation Model

The core innovation of this system is the aggregation of multiple receiver nodes into a unified power output. The total system power (P_{total}) is defined as the summation of the output of each node in the array:

$$P_{total} = \sum_{i=1}^{n} P_{node,i}$$

In this model, the system is designed to function as a singular, cohesive energy harvesting network, where $P_{node,i}$ represents the power captured by the i-th node.

4 System Potential

By scaling the number of nodes (n) and maintaining high-precision frequency synchronization, the system facilitates a scalable output. This design focuses on maximizing energy throughput by ensuring that every node acts as an efficient conduit for the targeted resonant energy, effectively transforming the grid into a high-capacity power harvester.